

Cavity Radar Cross Section Prediction

Adam Zdunek and Waldemar Rachowicz

Abstract—Alternative models are discussed for the determination of the interior irradiation contribution to the radar cross section (RCS) of open-ended cavities. Typical applications of practical interest include radiation field prediction of open-ended waveguides and signature prediction of jet engine air intakes and exhaust outlets. It is shown and explained why the classic perfectly conducting (PEC) ground plane (GP) model sometimes predicts measured cavity RCS very poorly. Examples where PEC GP model predictions are 5 dB or more off are provided. Verifications that the used PEC GP model is correctly implemented are provided. A physical optics based modification of the classic PEC GP model is provided and explained. A novel model where the cavity aperture forms an opening in an infinite radar absorbing (IBC) screen is proposed verified and validated. The IBC model resembles the common experimental set-up, and poses an efficient novel half-space type replacement model approximating a full space setting. The modified PEC GP model or the IBC model can be used interchangeably for cavities with electrically large apertures.

Index Terms—Absorbing screen, aperture integration, radar signature, radar cross section (RCS) prediction, reflecting screen.

I. INTRODUCTION

THE problem of reliably predicting the radar signature contribution of jet-engine air inlets and exhaust outlets to the total radar cross section (RCS) of an aircraft is one of the most challenging scattering problems in computational electromagnetics. The research effort spent on electromagnetic scattering analysis of open-ended cavities, waveguides, cavity backed aperture antennas, and ducts is profound, see the extensive review by Anastassiou [1]. This investigation is concerned with the prediction of the major, interior irradiation, contribution to the cavity RCS, hereafter denoted cavity RCS for short. That is, scattering contributions from the exterior structure, and coupling effects between the cavity and the structure exterior to the cavity are excluded. The interior cavity problem and the exterior scattering problem are fully coupled however. For a recent account including coupling with exterior structure contributions see [2]. Contributions from rim diffraction are often included in the cavity RCS and this is the case here too.

In passing it should be mentioned that experimental set-ups often use radar absorbing materials (RAM) to mask scattering

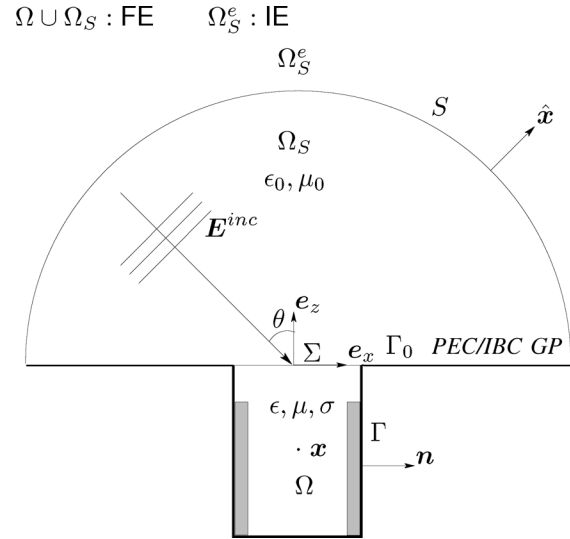


Fig. 1. A cavity backed aperture recessed in an infinite PEC/IBC ground-plane (GP).

on the exterior structure of the cavity, see [3], and references therein.

Prediction models of cavity RCS therefore do in one way or another exclude the scattering contribution from the exterior to the structure. In other words, they try to mimic the experimental set-ups.

Two major principally different approaches to predict cavity RCS are noted. Either the cavity is placed in free-space and the cavity RCS is computed taking only the scattering from the aperture into consideration, or the cavity opening is considered to form an aperture in a perfectly conducting (PEC) infinite ground plane (GP) which can be accounted for. The contribution to the scattered far-field from the exterior to the cavity is in both cases removed.

The PEC-ground plane approach, is well established and a popular approach, see Jin [4, Sec. 9.3], Volakis [5, Sec. 7.4] and Orfanidis [6, Ch. 16]. Cavity RCS predictions using the PEC GP model can be found in Jin [7] and Liu and Jin [8] using finite element-boundary integral equation based approaches, or with pure boundary integral approaches in Ammari *et al.* [9] and in Xuo [10], or by the present authors using a finite element-infinite element approach [11]–[13]. The scattered far-field is in the PEC GP approach obtained by an exact integral formula given the effective magnetic current on the aperture.

This investigation started as a consequence of the discovery that the PEC ground plane model predicted measured cavity RCS poorly for the well-known “simple jet engine” benchmark problem [14]–[16] and [17], see Fig. 8. A variant of the PEC GP model where the PEC screen is replaced by an absorbing (IBC) screen proved to predict the cavity RCS much better. The

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illumination and the far-field evaluation had of course to be adjusted appropriately for the absorbing screen set-up.

The absorbing screen concept is inspired by the experimental procedures using radar absorbents on the exterior of the cavity. A costly full space modelling with finite elements is avoided by using an infinite flat screen. The IBC GP model relies on approximating assumptions in order to justify an aperture based integration procedure to obtain the scattered far-field. The Kirchhoff type approximations that are invoked are expressed by Pathak and Burkholder [18] as follows: the excitation wavelength must be sufficiently short so that the aperture diameter becomes large compared to the wavelength, and, the incidence angle must not be close to grazing the aperture in order to avoid scattering on, i.e., illumination of, the exterior structure of the cavity, see Obelleiro-Basteiro *et al.* [19, Eqs. (1) and (11)], Xiang and Chia [20, Eq. (5)] and contributions by Burkholder *et al.* [21]–[24] and by Anastassiou [25], [15].

The question why the well established PEC GP model¹ produces poor cavity RCS predictions is also addressed. After a review of the PEC GP and IBC GP models we have an understanding involving the scattering contribution from the PEC aperture lid implied by the reflected field invoked in the PEC GP model. It fully explains the poor behavior of the classic PEC GP model in the cases studied.

We have not found any publication where cavity RCS predictions obtained with the PEC GP model are compared to actual measurements for a cavity representing an air-intake, exhaust outlet or alike. It is hard to see a practical use of this model in the indicated context if it cannot model real electrically large cavities. This investigation provides the missing comparison, it provides verifications of the implemented models, and it provides validations of predictions against measurements. Finally, we provide verifications that the proposed physical optics based modification to the PEC GP model works.

The outline of the study is as follows. The scattering from a cavity is described as a coupled exterior-interior problem in Section II where the used plane wave illumination, geometry, governing equations and boundary conditions are given. The post-processing steps to obtain the RCS are outlined in Section III including the involved aperture integration procedures to obtain the scattered far-field for the PEC and IBC ground plane models, respectively. Model verifications, validations and comparisons are made in Section IV using our own numerical results, published numerical results and published measurements on available generic cavities. The proposed modification to the PEC GP model is verified in Section V. The findings are discussed in Section VI and these are summarized and concluded in Section VII.

II. SCATTERING FROM A CAVITY

In this section we define the alternative boundary value problems used for cavity RCS determination in connection to our finite element – infinite element (FE-IE) approach.

Here, we present the formulation for the plane wave illumination of a cavity with perfectly conducting (PEC) walls whose

¹as defined by Jin [4, Sec. 9.3, Eq. (9.112)] using an EFIE formulation, and coherently by Volakis *et al.* [5, Sec. 7.4, Eqs. (7.43)+(7.21)] using a MFIE formulation

opening forms an aperture in an infinite flat screen, hereafter denoted ground plane (GP), as shown in Fig. 1. The screen will be either PEC, or it will be absorbing, approximated by an impedance boundary condition (IBC). In other words, we consider two alternative formulations, one with a PEC ground plane and the other with an IBC ground plane. We seek the scattered field in the half-space above the screen. Using a domain decomposition formulation the cavity domain below the aperture is common to both formulations while the exterior domain differs only with respect to the boundary condition representing the screen. We note that both types of RCS characterizations can be efficiently determined using a single mesh by forming separate boundary impedance operators for the exterior.

We consider time-harmonic electromagnetic fields with circular frequency $\omega (> 0)$ and denote the imaginary unit by $j (= \sqrt{-1})$. The common time dependence specified by the factor $\exp(j\omega t)$ will be omitted henceforth.

The cavity domain is henceforth denoted Ω . The cavity walls Γ are perfectly conducting, possibly coated by a layer of radar absorbing material (RAM).

The exterior to the cavity is taken to be the upper half-space, $\mathbb{R}_+^3 = \{\mathbf{x} \in \mathbb{R}^3 : z > 0\}$, bounded below by the ground plane $\Gamma_0 = \{\mathbf{x} \in \mathbb{R}^3 : z = 0\}$. The cavity opening Σ forms an aperture in Γ_0 , that is, $\Sigma \subset \Gamma_0$. An auxiliary truncating spherical surface $S = \{\mathbf{x} \in \mathbb{R}^3 : \|\mathbf{x}\|^2 = R^2, z > 0\}$ is introduced to aid the analysis. The upper half-space $\mathbb{R}_+^3$ is partitioned into a bounded domain denoted $\Omega_S = \{\mathbf{x} \in \mathbb{R}^3 : \|\mathbf{x}\|^2 < R^2, z > 0\}$ with boundary $\partial\Omega_S = S \cup \Gamma_0$, and the exterior Ω_S^c .

As mentioned in the Introduction there are two major principally different approaches to predict cavity RCS. Below we elaborate on the connection between them.

A. Model Problems for Cavity RCS

Consider scattering of an incoming wave $\mathbf{E}^i$ on two different geometries:

- (1) a bounded PEC cavity with non-scattering exterior immersed in a 3D space;
- (2) the same cavity grounded in an infinite PEC plane, cf. Fig. 1.

The main question is; can the solution to problem (1) be constructed from the solution of problem (2), and vice versa?

In general the induced electric currents on the cavity walls Γ will not be equal in problem (1) and problem (2), i.e., $\mathbf{J}_1|_\Gamma \neq \mathbf{J}_2|_\Gamma$ due to the coupling with the PEC plane $\Gamma_0 \setminus \Sigma$ in problem (2). This is readily understood taking into consideration the corner effects close to the aperture in problem (2). Therefore, there is no way in general to relate the scattered field solutions $\mathbf{E}_1^{\text{sc}}(\mathbf{x})$ and $\mathbf{E}_2^{\text{sc}}(\mathbf{x})$ in the upper half-space $\mathbb{R}_+^3$ to these two problems.

An approximate computable relation can be set-up, however, assuming that

- (i) the coupling between the electric current $\mathbf{J}_2|_\Gamma$ induced on the cavity walls and the electric current $\mathbf{J}_2|_{(\Gamma_0 \setminus \Sigma)}$ induced on the PEC plane $\Gamma_0 \setminus \Sigma$ in problem (2) can be neglected;
- (ii) the electric current $\mathbf{J}_2|_{(\Gamma_0 \setminus \Sigma)}$ on the PEC plane can be approximated as being uniform, corresponding to the reflection of incoming field $\mathbf{E}^i$ from infinite PEC surface Γ_0 , though it vanishes over the aperture Γ_0 .

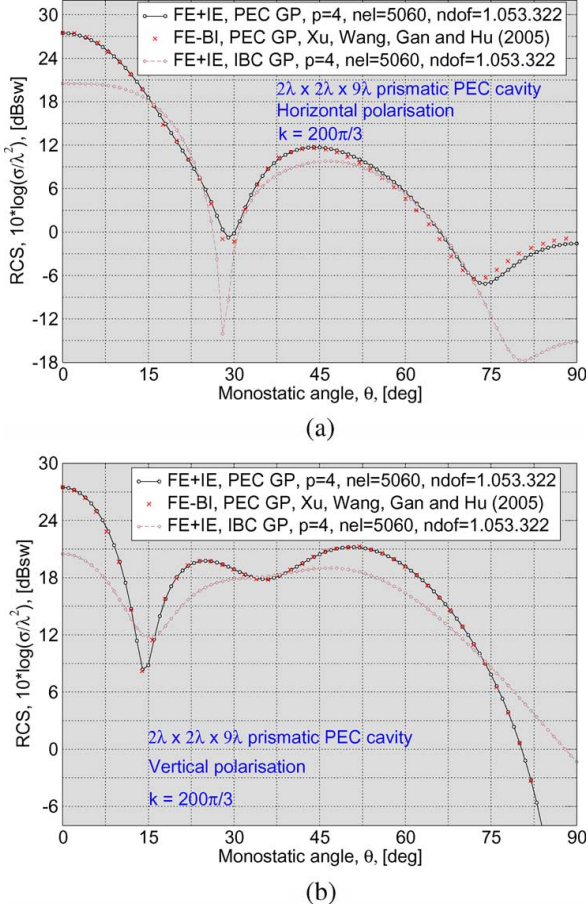


Fig. 2. Prismatic PEC $2\lambda \times 2\lambda \times 9\lambda$ cavity. Monostatic RCS versus azimuth angle θ at 10 GHz. Comparison with FE-BI results *with* PEC ground plane from Xu [10]. (a) Horizontal polarization. (b) Vertical polarization.

Accepting these two hypotheses we can use the superposition principle to express $\mathbf{E}_1^{\text{sc}}(\mathbf{x})$ by $\mathbf{E}_2^{\text{sc}}(\mathbf{x})$ for $\mathbf{x} \in \mathbb{R}_+^3$

$$\begin{aligned} \mathbf{E}_1^{\text{sc}} &\approx \mathbf{E}_2^{\text{sc}} - \mathbf{E}(\mathbf{J}\chi_{(\Gamma_0 \setminus \Sigma)}) \\ &= \mathbf{E}_2^{\text{sc}} - \mathbf{E}(\mathbf{J}\chi_{(\Gamma_0)}) + \mathbf{E}(\mathbf{J}\chi_{(\Sigma)}) = \mathbf{E}_2^{\text{sc}} - \mathbf{E}^r + \mathbf{E}(\mathbf{J}\chi_{(\Sigma)}). \end{aligned} \quad (1)$$

Above $\mathbf{E}(\cdot)$ is a field generated by the argument electric current distributed over plane Γ_0 , χ_A is a characteristic function of set A , (i.e., $\chi_A(\mathbf{x}) = 1$ if $\mathbf{x} \in A$, $\chi_A(\mathbf{x}) = 0$ otherwise), and $\mathbf{E}^r$ is the out-going field reflected from the PEC plane Γ_0 *without aperture*. Therefore we have

$$\mathbf{E}_1^{\text{sc}} \approx \mathbf{E}_2^{\text{sc}} - (\mathbf{E}^r - \mathbf{E}(\mathbf{J}\chi_{(\Sigma)})) \quad (2)$$

where the difference within parenthesis approximates the induced electric field on the PEC plane *with aperture*. Thus the scattered field $\mathbf{E}_1^{\text{sc}}$ can be obtained approximately by subtracting from $\mathbf{E}_2^{\text{sc}}$ the specular reflection $\mathbf{E}^r$ in Γ_0 , and by adding the field scattered by the hypothetical PEC cover of the cavity $\mathbf{E}(\mathbf{J}\chi_{(\Sigma)})$.

Provided the aperture is electrically large, $\mathbf{E}(\mathbf{J}\chi_{(\Sigma)})$ can be approximated by physical optics rules and its contribution to

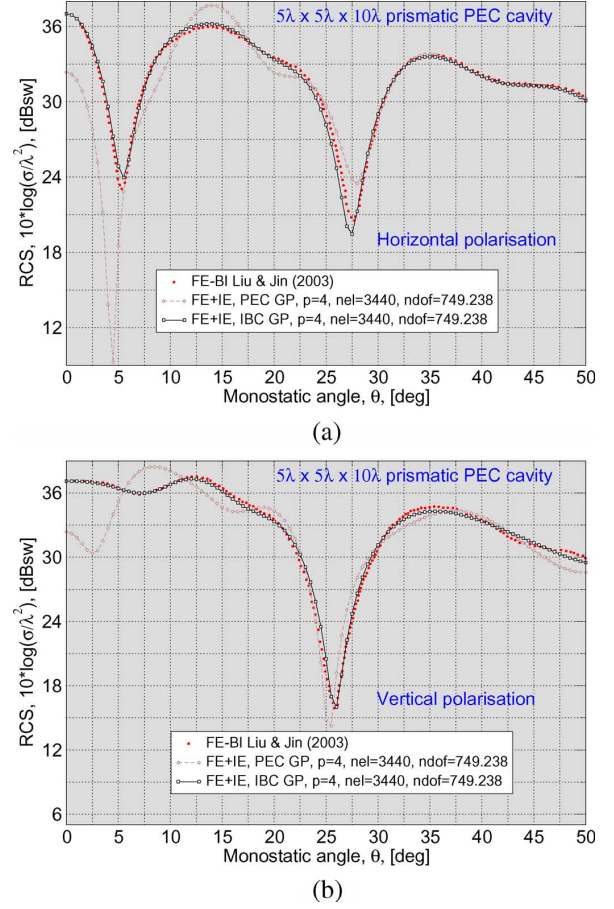


Fig. 3. Prismatic PEC $5\lambda \times 5\lambda \times 10\lambda$ cavity. Monostatic RCS versus azimuth angle θ at 0.3 GHz. Comparison with FE-BI results *without* a PEC ground plane from Liu and Jin [2]. (a) Horizontal polarization. (b) Vertical polarization.

the scattered field in the far zone can easily be computed, see Section V.

Note that by $\mathbf{E}_2^{\text{sc}}$ we understand the member of the usual additive splitting considered in scattering problems

$$\mathbf{E}_2^{\text{sc}} := \mathbf{E}_2^{\text{tot}} - \mathbf{E}^i \quad (3)$$

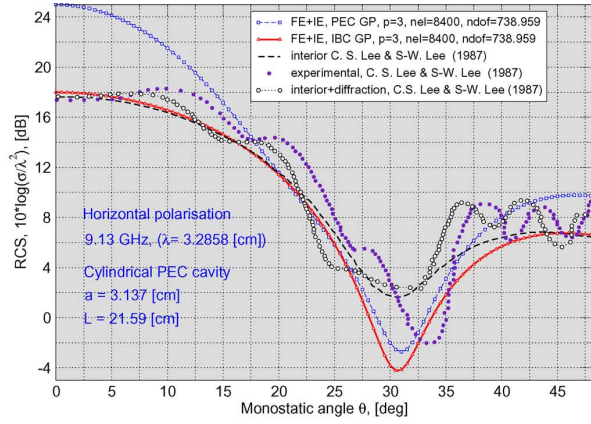
with $\mathbf{E}_2^{\text{tot}}$ being the total field satisfying the Maxwell equations and homogeneous Dirichlet boundary conditions on PEC surfaces. The difference field $\tilde{\mathbf{E}}_2^{\text{sc}} := \mathbf{E}_2^{\text{sc}} - \mathbf{E}^r$ in (2), is the part of $\mathbf{E}_2^{\text{sc}}$ with appropriate $1/r$ decay in the far zone. In terms of $\tilde{\mathbf{E}}_2^{\text{sc}}$ split (3) becomes

$$\tilde{\mathbf{E}}_2^{\text{sc}} = \mathbf{E}_2^{\text{tot}} - (\mathbf{E}^i + \mathbf{E}^r). \quad (4)$$

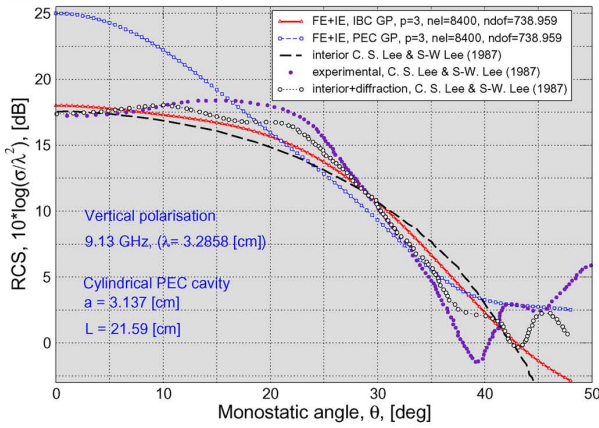
We conclude that we may solve problem (2) directly for $\tilde{\mathbf{E}}_2^{\text{sc}}$ by assuming the incident field as $\mathbf{E}^{\text{inc}} = \mathbf{E}^i + \mathbf{E}^r$. A scattered field formulation of problem (1) in terms of $\mathbf{E}_1^{\text{sc}}$ relies on the other hand, on the splitting

$$\mathbf{E}_1^{\text{sc}} = \mathbf{E}_1^{\text{tot}} - \mathbf{E}^i \quad (5)$$

where $\mathbf{E}_1^{\text{tot}}$ has the same meaning as before. The fact that $\tilde{\mathbf{E}}_2^{\text{sc}}$ can be determined directly is a lucky coincidence as the right



(a)



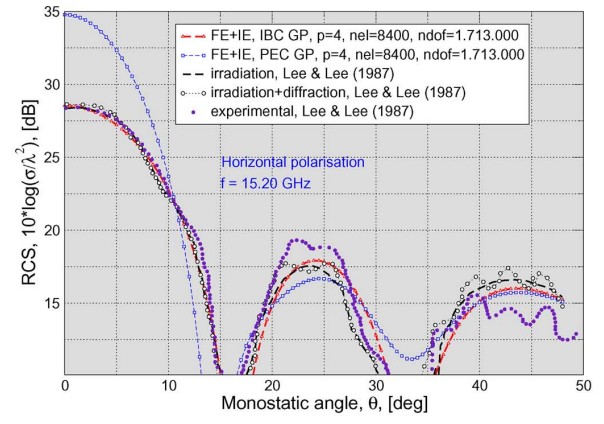
(b)

Fig. 4. Cylindrical PEC cavity, radius 3.137 cm, length 21.59 cm. Monostatic RCS versus azimuth angle θ at 9.13 GHz. (a) Horizontal polarization. (b) Vertical polarization.

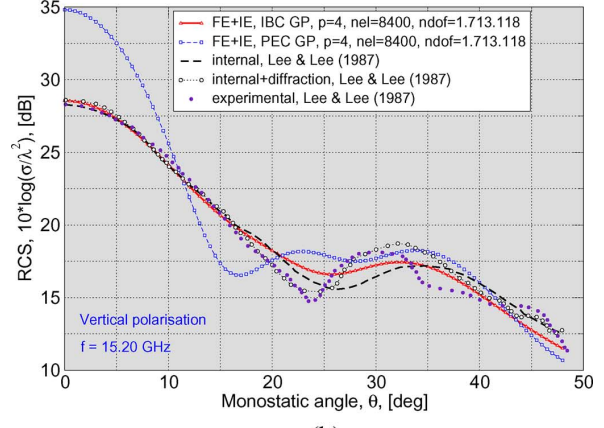
$1/r$ decay is a strict point-wise requirement of the Silver–Müller radiation condition and in a weak sense by the infinite elements in our finite element approach.

Finally, it is evident that a cavity RCS prediction determined with the PEC GP model [problem (2)] will differ from that determined using the free-space model [problem (1)]. The difference will depend on the magnitude of the far-field contribution from the lid scattering term $\mathbf{E}(\mathbf{J}_{\chi(\Sigma)})$. The discrepancy in predicted RCS will obviously increase with the size of the aperture. From physical optics we conclude that the contribution from $\mathbf{E}(\mathbf{J}_{\chi(\Sigma)})$ will depend on the direction $\mathbf{d}^i$ of the illuminating field as $e^{-jk_0(\mathbf{d}^i \cdot \mathbf{t})}$ where $\mathbf{t}$ is tangent to the aperture. That is, its contribution will be largest for normal incidence and it will decline exponentially as the angle of incidence departs from normal.

Remark 1 (Reflected Field Contribution): In the finite element boundary integral version of the PEC GP model as defined by Jin in [4, Sec. 9.3] the reflected field enters the model through the use of the half-space Green's function of the electric type satisfying the Silver–Müller radiation condition and the PEC boundary condition $\mathbf{n} \times \mathbf{G}_e = \mathbf{0}$ at the GP, i.e., on the whole Γ_0 . The out-going field $\mathbf{E}^r$ is identified as the reflected field generated by the image source to the primary source being responsible for the incoming field $\mathbf{E}^i$. Jin [4, Eq. (9.112)] and our



(a)



(b)

Fig. 5. Cylindrical PEC cavity, radius 3.137 cm, length 21.59 cm. Monostatic RCS versus azimuth angle θ at 15 GHz. (a) Horizontal polarization. (b) Vertical polarization.

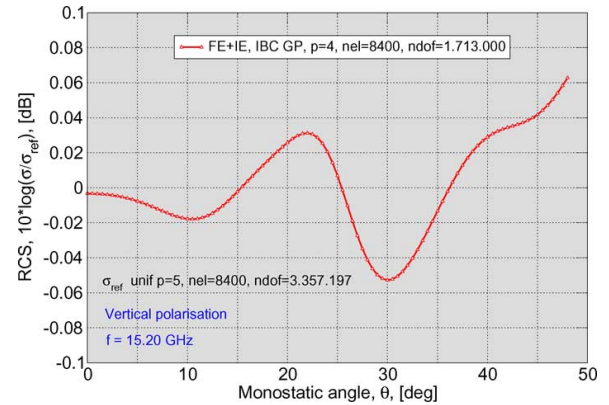


Fig. 6. Cylindrical PEC cavity, radius 3.137 cm, length 21.59 cm. Convergence assessment. Difference in monostatic RCS versus azimuth angle θ . Uniform $p = 5$ extensions with 3.3 MDof used as reference solution. Vertical polarization. 15.20 GHz.

(4) are equivalent splits. Confer also with Volakis *et al.* [5, Sec. 7.4, Eqs. (7.43)+(7.21)] who uses an equivalent MFIE based definition.

In other words, our FE+IE based formulation of the classic PEC GP model, based on (8b), (9a)–(9j), and (10b) below, is coherent with those given in [4] and [5]. ■

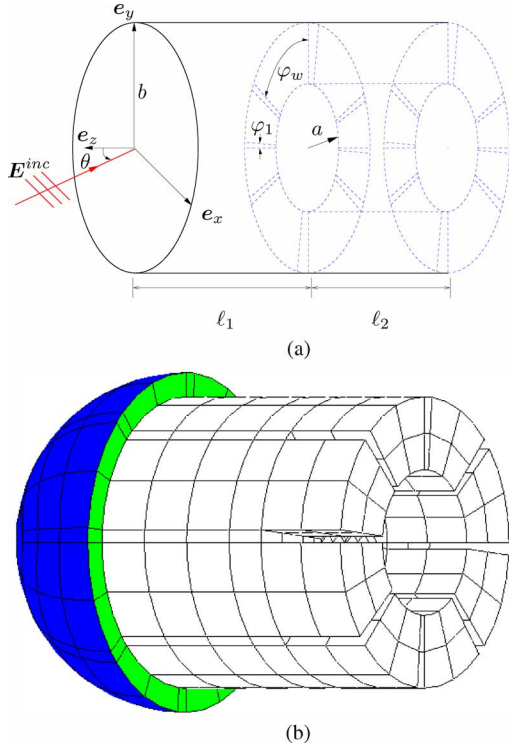


Fig. 7. (a) Geometry for the simple jet engine model. Straight bladed termination. (b) Mesh and boundary condition mark-up. IBC datum plane model. White = PEC, Black = Infinite Elements (IE), Gray = Impedance Boundary Conditions.

We can now proceed to set up our model problems. The incoming field is specialized to be a linearly polarized plane wave on the form

$$\mathbf{E}^i = \mathbf{p}^i \exp(-jk \mathbf{d}^i \cdot \mathbf{x}) \quad (6)$$

where $k = \omega \sqrt{\epsilon \mu}$ is the wave number, wherein ϵ is the (electric) permittivity, μ is the (magnetic) permeability, $\mathbf{d}^i$ is a unit vector giving the propagation direction and $\mathbf{p}^i$ is a unit vector telling the polarization, such that $\mathbf{p}^i \cdot \mathbf{d}^i = 0$, $\mathbf{d}^i \cdot \mathbf{e}_z < 0$. Given $k, \mathbf{d}^i$ and $\mathbf{p}^i$ the plane wave $\mathbf{E}^r$ reflected in Γ_0 PEC becomes

$$\mathbf{d}^r = \mathbb{R} \mathbf{d}^i, \quad \mathbf{E}^r = -\mathbb{R} \mathbf{E}^i \quad (7)$$

where the reflection dyadic for Γ_0 PEC is $\mathbb{R} = \mathbb{I} - 2\mathbf{e}_z \otimes \mathbf{e}_z$. So determined, the sum $\mathbf{E}^i + \mathbf{E}^r$ satisfies homogeneous Maxwell's equations and the PEC boundary condition $\mathbf{e}_z \times (\mathbf{E}^i + \mathbf{E}^r) = \mathbf{0}$ on Γ_0 .

Summing up, in our scattering problems the incident field $\mathbf{E}^{\text{inc}}(\mathbf{x}) = \mathbf{E}^{\text{tot}}(\mathbf{x}) - \mathbf{E}^{\text{sc}}(\mathbf{x})$ is assumed on the form

$$\mathbf{E}^{\text{inc}} = \begin{cases} \mathbf{E}^i, & \text{(IBC GP)} \\ \mathbf{E}^i + \mathbf{E}^r, & \text{(PEC GP)} \end{cases} \quad (8a) \quad (8b)$$

for the IBC and PEC GP models, respectively. It should be noted that the IBC GP model effectively belongs to the first enumerated class of cavity scattering problems considered on the previous page.

We now discuss the coupled Maxwell problem governing the scattering on a cavity recessed in a perfectly conducting, or al-

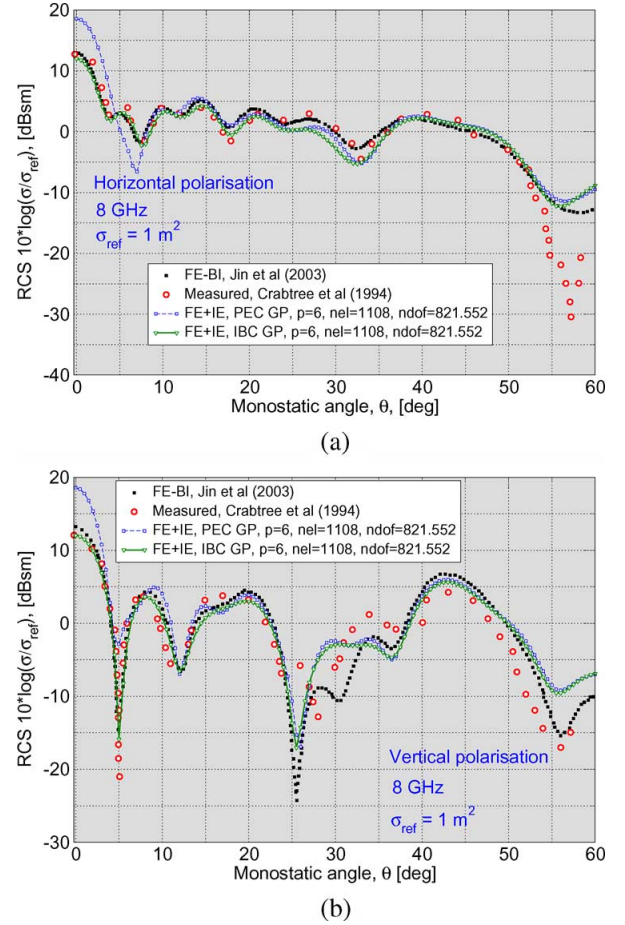


Fig. 8. Simple model of jet engine air-intake. Monostatic RCS versus azimuth angle θ at 8 GHz. (a) Horizontal polarization. (b) Vertical polarization.

ternatively, in an absorbing (IBC) ground plane. Towards this goal, we split the computational domain into three parts: (a) far-field domain Ω_S^e , (b) near-field domain Ω_S , and the cavity domain Ω . We set to determine: (a) scattered field $\mathbf{E}_1$ defined in Ω_S^e , total field $\mathbf{E}_2 = \mathbf{E}^{\text{inc}} + \mathbf{E}^{\text{sc}}$ defined in Ω_S , and the total field $\mathbf{E}_3 = \mathbf{E}^{\text{inc}} + \mathbf{E}^{\text{sc}}$ defined in Ω .

The coupled problem looks as follows:

$$\begin{cases} \nabla \times \nabla \times \mathbf{E}_1 - k_0^2 \mathbf{E}_1 = \mathbf{0} & \text{in } \Omega_S^e & (9a) \\ r[\nabla \times \mathbf{E}_1 - jk_0 \hat{\mathbf{x}} \times (\hat{\mathbf{x}} \times \mathbf{E}_1)] \rightarrow \mathbf{0} & \text{as } r \rightarrow \infty & (9b) \\ \nabla \times \nabla \times \mathbf{E}_2 - k_0^2 \mathbf{E}_2 = \mathbf{0} & \text{in } \Omega_S & (9c) \\ \mathbf{e}_z \times (\mathbf{E}_2 \times \mathbf{e}_z) + \eta Z_0 \mathbf{e}_z \times \mathbf{H}_2 = \mathbf{0}, & \text{on } \Gamma_0 \setminus \Sigma & (9d) \\ \nabla \times (\mu_3^{-1} \nabla \times \mathbf{E}_3) - \omega^2 \bar{\epsilon}_3 \mathbf{E}_3 = \mathbf{0} & \text{in } \Omega & (9e) \\ \mathbf{n} \times \mathbf{E}_3 = \mathbf{0} & \text{on } \Gamma & (9f) \\ \hat{\mathbf{x}} \times \mathbf{E}_2 = \hat{\mathbf{x}} \times (\mathbf{E}_1 + \mathbf{E}^{\text{inc}}), & \text{on } S & (9g) \\ \hat{\mathbf{x}} \times \mathbf{H}_2 = \hat{\mathbf{x}} \times (\mathbf{H}_1 + \mathbf{H}^{\text{inc}}), & \text{on } S & (9h) \\ \mathbf{e}_z \times \mathbf{E}_2 = \mathbf{e}_z \times \mathbf{E}_3 & \text{on } \Sigma & (9i) \\ \mathbf{e}_z \times \mathbf{H}_2 = \mathbf{e}_z \times \mathbf{H}_3 & \text{on } \Sigma. & (9j) \end{cases}$$

Here $\hat{\mathbf{x}}$ denotes the exterior unit normal to sphere S , pointing into domain Ω_S^e , η is a normalized surface impedance, and $\bar{\epsilon}_3 = \epsilon_3 + \sigma_3/j\omega$, wherein σ denotes the conductivity. Note that the

right hand side of (9a) expands to $\mathbf{0} = k_0^2(\bar{\epsilon}_1/\epsilon_0 - \mu_0/\mu_1)\mathbf{E}^{\text{inc}}$ since $\bar{\epsilon}_1 = \epsilon_0 = \mu_0 = \mu_1 = 1$.

We consider the two cases

$$\begin{cases} \eta \neq 0, & \text{on } \Gamma_0 \setminus \Sigma \text{ (IBC GP)} \\ \eta = 0, & \text{on } \Gamma_0 \setminus \Sigma \text{ (PEC GP)}. \end{cases} \quad (10a)$$

$$(10b)$$

The solution to the ultimate coupled problem is formed by enforcing electromagnetic field continuity across the domain boundaries S and Σ using (9g)–(9j). Also note that the illumination comes in via (9g) and (9h). The coupled cavity scattering problem (9b)–(9j) is solved numerically using the finite element–infinite element based approach described in [12] and more recently in [26]. Finite elements discretize Ω and Ω_S , while infinite elements cover Ω_S^c . We use the domain decomposition (DD) procedure described in detail in Demkowicz *et al.* [26]. It allows us to determine the RCS characterizations for the PEC and IBC models efficiently using a single mesh by forming separate boundary impedance operators for the exterior. Finally, it should be noted that a scattered field formulation must be used in domain Ω_S^c due to the radiation condition (9b) whereas we are in principle free to use either a total field formulation or a scattered field formulation in domains Ω and Ω_S .

III. RADAR CROSS SECTION (RCS) DETERMINATION

Needless to say, the quantity of interest in radar signature prediction of cavities is the radar cross section (RCS) defined as

$$\begin{aligned} \sigma(\hat{\mathbf{x}}, \hat{\mathbf{d}}^i) &= \lim_{r \rightarrow \infty} 4\pi r^2 \frac{1/2\Re\{\mathbf{E}^{\text{sc}} \times \bar{\mathbf{H}}^{\text{sc}}\} \cdot \hat{\mathbf{x}}}{1/2\Re\{\mathbf{E}^i \times \bar{\mathbf{H}}^i\} \cdot \hat{\mathbf{d}}} \\ &= \lim_{r \rightarrow \infty} 4\pi r^2 \frac{\|\mathbf{E}^{\text{sc}}\|^2}{\|\mathbf{E}^i\|^2} \end{aligned} \quad (11)$$

expressing the power flow density (per unit area) carried by the scattered field $\mathbf{E}^{\text{sc}}$ away from the scatterer in the direction $\hat{\mathbf{x}} = \mathbf{x}/r$ with $r = |\mathbf{x}|$, relative to the power flow density carried by the $\hat{\mathbf{d}}^i$ -directed incoming field $\mathbf{E}^i$. In (11) r is the radius of the sphere where the scattered field is evaluated.

In the engineering literature, RCS is often given as a dimensionless quantity $10 \cdot \log(\sigma/a)$, where a is a reference area. We adhere to the engineering definition with the associated units [dBsm], and [dBsw] corresponding to the normalization by 1 m^2 , and by the actual wavelength squared λ^2 , respectively.

Our finite element computations can only provide the scattered field $\mathbf{E}^{\text{sc}}$ at a finite distance from the cavity opening. The asymptotic behavior in the far-field zone is requested in the RCS definition (11). This behavior is determined next.

A. The Resulting Far-field

We recall the Stratton–Chou integral equation expression for the scattered electromagnetic fields as a function of the equivalent (effective) electric and magnetic currents $\mathbf{J} = \mathbf{e}_z \times \mathbf{H}$ and $\mathbf{M} = \mathbf{E} \times \mathbf{e}_z$ on the aperture Σ of the cavity for the PEC and IBC ground plane cases, respectively. In the far-field limit these

formulas yield the following common expression for the scattered electric field

$$\begin{aligned} \mathbf{E}^{\text{sc}}(\mathbf{x}) &\sim jk \frac{e^{-jkr}}{4\pi r} \mathbf{F}(\hat{\mathbf{x}}) \quad \text{as } r \rightarrow \infty \\ \mathbf{F}(\hat{\mathbf{x}}) &= \int_{\Sigma} [Z\hat{\mathbf{x}} \times (\hat{\mathbf{x}} \times \alpha \mathbf{J}(\mathbf{y})) + \hat{\mathbf{x}} \times \beta \mathbf{M}(\mathbf{y})] e^{jk(\hat{\mathbf{x}} \cdot \mathbf{y})} dA \end{aligned} \quad (12)$$

see, [4], [5], [26]. The appropriate relationship is specified by assigning the parameters α and β for the actual case

$$\begin{cases} \alpha = 1, \beta = 1, & \text{(IBC GP)} \\ \alpha = 0, \beta = 2, & \text{(PEC GP)} \end{cases} \quad (13)$$

The formula for the PEC ground plane case is *exact*, confer derivations in Jin [4, Sec. 9.3] and in Demkowicz *et al.* [26, Sec. 11.4].

The resulting expression for the (IBC GP) case is known as *Kirchhoff's aperture integration formula*. It is *approximate* and it is developed for scattering by a body in free-space ($\mathbb{R}^3$) under the assumption that the scattering contribution from the aperture is dominating.

The approximations invoked by the Kirchhoff procedure are as follows:

- the excitation wavelength λ must be sufficiently short so that the aperture diameter becomes large relative to λ ;
- the exciting wave must not have an incidence angle which departs too much from normal incidence in order not to illuminate the exterior structure.

Evidently, for a fixed channel geometry, this approximation will be justified for high frequencies and with radar absorbing material (RAM) coated exteriors that do not contribute to the scattering. The fact that the simple absorbing boundary condition is approximate, and that its efficiency deteriorates with incidence angles departing from normal is noted. The essence of Kirchhoff's approximation is the assumption that the scattered field can be additively split into an aperture contribution, and a contribution from the remainder [18]. Applying it to the IBC GP model we obtain

$$\mathbf{E}^{\text{sc}}(\mathbf{x}) = \mathbf{E}_{\Sigma}^{\text{sc}}(\mathbf{x}) + \underbrace{\mathbf{E}_{\Gamma_0 \setminus \Sigma}^{\text{sc}}(\mathbf{x})}_{\approx 0} \quad (14)$$

where the second member can be neglected if the scattering contribution from the absorbing screen, $\Gamma_0 \setminus \Sigma$ is small compared to the contribution from the aperture.

Remark 2 (PEC GP Model Adjustment): According to (2) we can approximately retrieve the scattered electric field in the upper half-space for a model of a cavity in free-space, i.e., *without* a PEC GP and with no contributions from the exterior, by taking into consideration the uniform electric current induced by the incident field on a hypothetical PEC aperture cover. That is

$$\mathbf{J} = \mathbf{e}_z \times \mathbf{H}^{\text{inc}}, \quad \text{on } \Sigma. \quad (15)$$

Starting from Faraday's law $-jk_0 Z_0 \mathbf{H}^{\text{inc}} = \nabla \times \mathbf{E}^{\text{inc}}$ and (8b) we can express the incident magnetic field as

$$\mathbf{H}^{\text{inc}} = 2(\mathbb{I} - \mathbf{e}_z \otimes \mathbf{e}_z) \mathbf{H}^i \quad (16)$$

where $Z_0 \mathbf{H}^i = \mathbf{d}^i \times \mathbf{E}^i$ wherein $\mathbf{E}^i = E_0 \mathbf{p}^i e^{-jk_0(\mathbf{d}^i \cdot \mathbf{x})}$ is the illuminating incoming plane wave. Using (16) in (15) we finally get

$$\mathbf{J} = \mathbf{e}_z \times 2\mathbf{H}^i \quad \text{on} \quad \Sigma. \quad (17)$$

That is, the electric current on the aperture we have to adjust for is as expected given by the physical optics (PO) expression (17).

The physical optics adjusted PEC GP model corresponds to using the parameter combination ($\alpha = 1, \beta = 2$) with $\mathbf{J}$ in (12) computed using (17). ■

IV. MODEL VERIFICATIONS, COMPARISONS AND VALIDATIONS

We first verify that the finite element implementations of the two comparison cavity RCS models reproduce published numerical results sufficiently well. Simultaneously, the PEC and IBC ground plane model predictions are compared so that the reader can see the substantial discrepancy in RCS prediction. We then compare our PEC and IBC ground plane model predictions with available RCS measurements, showing that the classic PEC GP model predicts cavity RCS poorly, and that the IBC model performs much better, thus rejecting the former while corroborating validation of the latter. Finally, we provide RCS predictions using the PEC GP model results adjusted for the scattering contribution originating from the virtual PEC lid over the aperture by using physical optics.

We start verifying that our finite element – infinite element implementation of the PEC ground plane model can reproduce published PEC GP predictions. Needless to say, verification is a necessary step preceding validation.

A. Prismatic $2\lambda \times 2\lambda \times 9\lambda$ PEC Cavity

We use the $2\lambda \times 2\lambda \times 9\lambda$ prismatic PEC cavity studied by Xu *et al.* [10] using a hybrid finite element integral equation based approach with a PEC GP to verify our FE+IE implementation of the PEC GP model. The hexahedral element mesh used consists of totally 5060 4th order ($p = 4$) elements for $\Omega \cup \Omega_S$. An analysis with uniform $p = 4$ approximation yields 1.053.322 degrees-of-freedom. Monostatic RCS is computed from $\theta = 0$ to 90 degrees with step-size $\Delta\theta = 1$ degrees. Our PEC GP based cavity RCS predictions in Fig. 2 are evidently in excellent agreement with those reported by Xu *et al.* [10].

In order to provide a self-contained verification of the implementation of the PEC ground plane model an additional example taken from a separate source is supplied in the Appendix A.

Clearly, these two examples show that our implementation of the PEC ground plane model is correct. For further verifying examples, see [26] and references therein.

Next we verify that the proposed IBC ground plane model is correctly implemented.

B. Rectangular $5\lambda \times 5\lambda \times 10\lambda$ PEC Cavity

We choose the rectangular $5\lambda \times 5\lambda \times 10\lambda$ PEC cavity studied by Liu and Jin [2] using the MFLMA and FE-BI methods as the main verification example of the IBC ground plane model. It should be noted that Liu and Jin use a full space ($\mathbb{R}^3$) setting in this case.

Fig. 3 shows that the IBC GP cavity RCS predictions obtained with our FE+IE formulation are in excellent agreement with those obtained by Liu and Jin [2] using a hybrid finite element – boundary integral based full space model, i.e., *without* a PEC GP.

A supplementary example using an approximate iterative physical optics (IPO) method corroborating the verification of the IBC model is provided in Appendix B. Further verifications of the proposed IBC model are provided in conjunction with the validation of the model below.

Finally, we note the large discrepancies between the PEC and the IBC model cavity RCS predictions displayed in Figs. 2 and 3. Most importantly, we note that the discrepancies are largest for acute angles of incidence.

We have verified our implementations of both models and we have pointed out the discrepancy. It remains to validate the proposed IBC GP model, and to explain the source of the large discrepancy between the comparison models.

We start by validating the proposed IBC GP model using measurements reported in Lee and Lee [27].

C. The Lee [27] Cylindrical Cavity

Lee and Lee report mode matching based predictions and measurements for a circular cylindrical PEC cavity of length 21.59 cm and radius 3.137 cm, terminated with a flat disc. We use a mesh consisting of 8400 3rd order ($p = 3$) hexahedral elements for $\Omega \cup \Omega_S$ in the simulation. So called exact geometry elements with cylindrical mapping are used for all element faces on the cylinder mantle while a spherical mapping is used for those located on the truncating exterior half-sphere covered by infinite elements. Figs. 4 and 5 present our monostatic RCS predictions for horizontal and vertical polarizations at 9.13 and 15.20 GHz, respectively, together with the numerical results and measurements due to Lee and Lee [27].

It should be noted that our finite element – infinite element approach includes diffraction contributions at the aperture edges both for the infinite PEC ground plane model, and for the IBC ground plane model. In the former we resolve the geometric edge singularity formed at the aperture rim, and in the latter case we deal with a combined geometric and material singularity due to the PEC-to-IBC jump at the aperture rim.

Finally, in Fig. 6 we present a simple convergence assessment study where the difference between the uniform $p = 4$ prediction of RCS and a reference prediction obtained on a uniform $p = 5$ mesh with 3.3 million degrees-of-freedom is plotted versus azimuth angle. The so estimated discretization error is less than 0.06 dB over the whole range of incidence angles.

D. A Simple Model of a Jet Engine Air Intake

In the second validation example we present RCS predictions for a generic model of a jet-engine air-intake first studied by

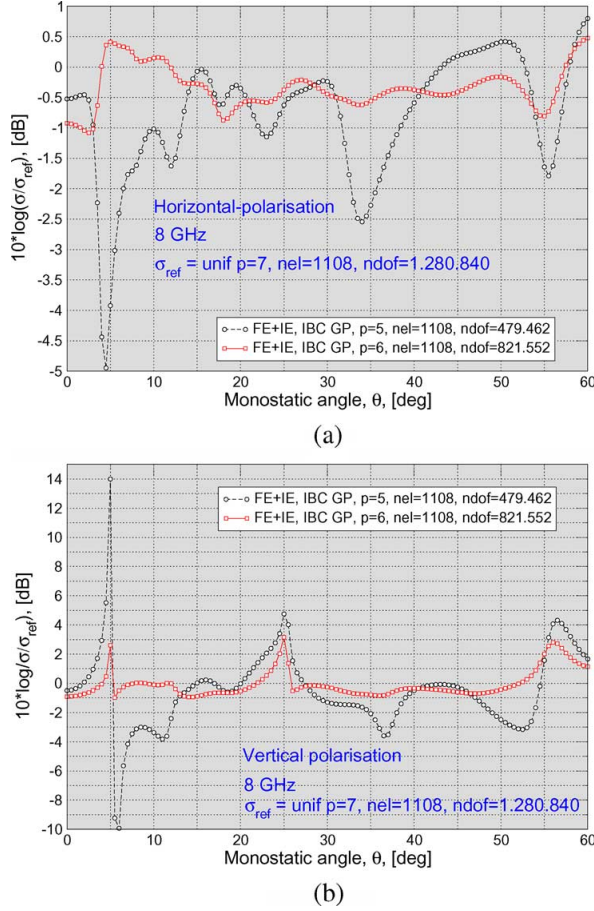


Fig. 9. Simple model of jet engine air-intake. RCS error relative 1.3 MdoF $p = 7$ solution versus azimuth angle θ at 8 GHz. (a) Horizontal polarization. (b) Vertical polarization.

Anastassiou [15], then by Jin *et al.* [16], and currently used as a benchmark problem in the NATO-SET085 research task group [17].

The model considered here is a straight circular cylindrical air-intake channel with a simple fan with eight straight blades mounted on a cylindrical hub, as depicted in Fig. 7(a). The cavity is terminated by a flat disk. The channel wall, the fan and the terminating disk are all considered being PEC. The radius of the hollow cylindrical section is $b = 0.15$ m and the length $\ell_1 = 0.20$ m. The length of the termination is $\ell_2 = 0.10$ m and the radius of the cylindrical hub $a = 0.075$ m. There are eight blades of angular thickness $\varphi_1 = 5^\circ$ and the separation angle $\varphi_w = 40^\circ$. The mid-plane of the first pair of blades is rotated 2.5° relative to the xz -plane. The mesh consisting of 1108 6th order ($p = 6$) hexahedral elements, for domains $\Omega \cup \Omega_S$ and boundary conditions are shown in Fig. 7(b). The mesh is truncated using infinite elements on a mapped spherical surface. The radius R of the truncating sphere S is 0.175 m. Hex surfaces adjacent to the cylindrical PEC mantle are mapped onto the cylindrical surface. The mapping allows us to use large elements.

Our RCS predictions for the infinite PEC GP model and the IBC GP model at 8 GHz are shown in Fig. 8. It is recalled that the same mesh, p -extension and domain decomposition is used for both simulations. Only the boundary conditions on the plane Γ_0 ($z = 0$) are different, see Fig. 1. These results are

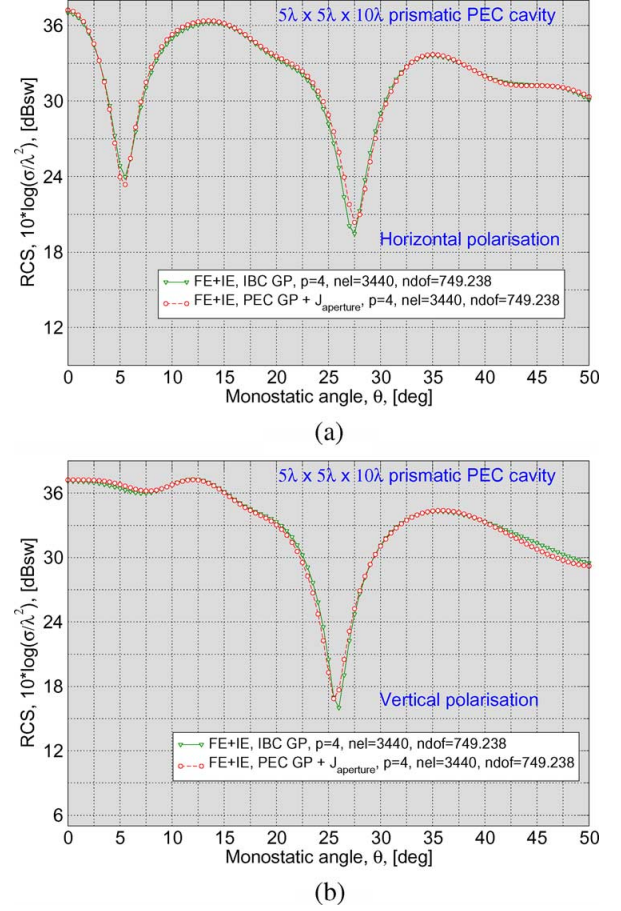


Fig. 10. Prismatic PEC $5\lambda \times 5\lambda \times 10\lambda$ cavity. Monostatic RCS versus azimuth angle θ at 0.3 GHz. Augmented PEC GP model using a PO based PEC aperture lid scattering contribution. (a) Horizontal polarization. (b) Vertical polarization.

compared to those obtained by Jin *et al.* [16] using a finite element boundary integral formulation (FE-BI), and to the measurements performed by Crabtree *et al.* [14]. In passing it should be mentioned that the findings in Fig. 8 started this investigation.

Provided RCS predictions are assessed to be reasonably converged using uniform $p = 5, 6$ -extensions compared to a $p = 7$ overkill solution, see Fig. 9.

V. A PO AUGMENTED PEC GP MODEL FOR CAVITY RCS PREDICTION

In Section IV-A we first verified that our implementation of the classic PEC GP model is correct. In Section IV-C and Section IV-D we then showed that it predicts measured cavity RCS poorly, especially in the important sector of acute angle incidence. We also showed that the proposed IBC GP model produces much better cavity RCS predictions coherent with published predictions using free-space models. It would be of great interest to be able to adjust the classical PEC GP model to produce cavity RCS predictions coherent with models not involving a PEC ground plane. Equation (2) suggests that the scattering contribution from the scattered field in the closed PEC aperture is likely to be responsible for the discrepancy between the PEC GP and IBC GP cavity RCS predictions. Here we show that this hypothesis holds truth and we verify that the

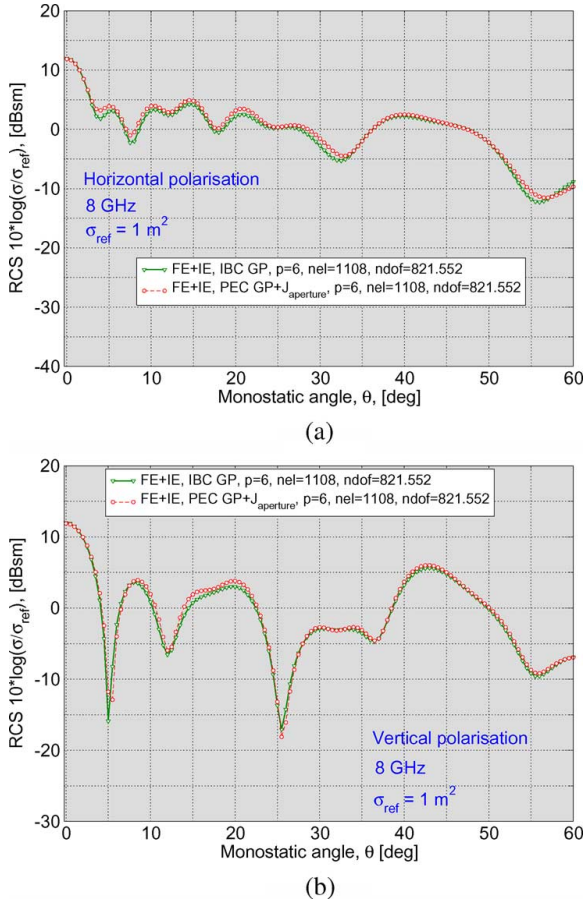


Fig. 11. Simple model of jet engine air-intake. Monostatic RCS versus azimuth angle θ at 8 GHz. PEC GP model adjustment using a PEC aperture lid scattering contribution. (a) Horizontal polarization. (b) Vertical polarization.

proposed adjustment to the PEC GP model defined in Remark 2 works.

Fig. 10 shows the adjusted PEC GP cavity RCS predictions relative to the IBC GP predictions using the Remark 2 recipe for the prismatic $5\lambda \times 5\lambda \times 10\lambda$ PEC cavity.

Finally, Fig. 11 shows the adjusted PEC GP cavity RCS predictions relative to the IBC GP predictions for the simple engine model at 8 GHz. Figs. 10 and 11 corroborate our hypothesis expressed by (2) fully. It should be recalled, however, that the hypothesis assumes that the conditions for the Kirchhoff approximation are met and that the diffraction on the aperture edges can be neglected.

VI. DISCUSSION

This investigation started by the discovery that our finite element–infinite element implementation of the well established PEC ground plane model described in [4, Sec. 9.3] and in [5, Sec. 7.4] predicts measured cavity RCS poorly for the well-known “simple jet engine” benchmark problem [14]–[16] and [17]. The findings are shown in Fig. 8. This finding was supported by the measurements reported by Lee [27] as shown in Section IV-C.

A variant of the PEC GP model replacing the PEC screen by an absorbing screen proved to predict the cavity RCS much

better as we have seen. The illumination and the far-field evaluation is of course adjusted appropriately for the absorbing screen set-up.

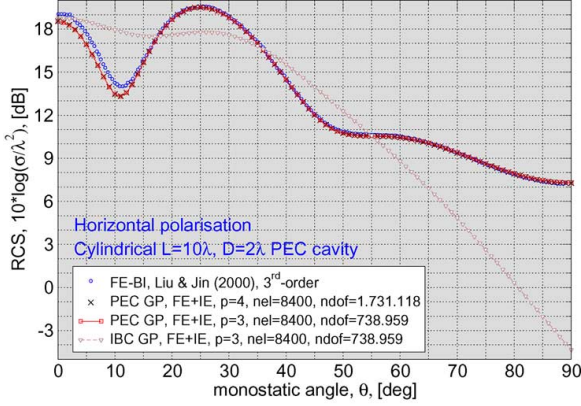
Revisiting the PEC GP model formulation a probable explanation to the behavior emerged and could be formulated.

In this article we present all elements that support our findings, we review the PEC and IBC model formulations (Section II), we point out the probable cause to the poor PEC GP model behavior and present arguments that the integral equation formulations of the PEC GP model in the textbooks [4, Sec. 9.3] and [5, Sec. 7.4] are equivalent to our finite element–infinite element formulation [Remark 2 and the motivation behind (2)], we then verify our implementation of the PEC and IBC models *per se* (Sections IV-A and IV-B, respectively), and we verify the aperture lid scattering adjustment proposed (Section V). We also validate the proposed IBC GP model and in retrospect also the PO adjusted PEC GP model against available measurements (Sections IV-C, IV-D and V) while showing the poor predictions of measured cavity RCS using the classic PEC GP model.

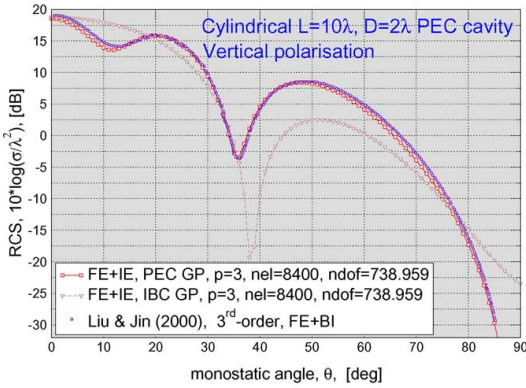
The objectives of this investigation are fully met but there are of course a number of issues worth investigating further. To mention a few, it is of interest to establish bounds within which the proposed PO correction of the PEC GP model works. The correction put forward relies on the reflected field to be uniform over the aperture (physical optics) which obviously is not true for cases where the edge diffraction on the aperture edges starts to matter. Investigation into better corrections not compromising the computational efficiency could be added to the wish list. If a halfspace Green’s function for the absorbing screen case could be constructed, we would not need to rely on the Kirchhoff approximation in the absorbing screen model. It would become exact in the same sense as the PEC GP model is. These additional issues fall outside the scope of this investigation, however, and will hopefully be dealt with elsewhere.

VII. SUMMARY AND CONCLUSION

It is shown that measured cavity RCS is sometimes very poorly predicted by the well established and popular PEC ground plane model. The discrepancy is shown to be worst for illuminations close to normal to the aperture, i.e., in the most critical sector. The cause to the poor modelling is identified and explained. Remedies are proposed. A direct physical optics based augmentation to the PEC ground plane model is provided, verified and validated. A replacement model to the classic PEC GP model is also provided, verified and validated. It is a variant, where the PEC GP is replaced by an absorbing screen, modelled by an impedance boundary condition (IBC). The illumination and the far-field evaluation procedure are appropriately adjusted for the IBC model. It is shown that the augmented PEC GP and the IBC GP models can be used interchangeably for electrically large apertures. Both the augmented PEC GP model and the IBC model are approximate. The IBC model relies on the Kirchhoff approximation, while the modification of the PEC GP model assumes the reflected field in a virtual PEC aperture lid to be uniform, i.e., the augmenting term is given by a physical optics expression. Replacing the IBC by a perfectly matched layer (PML) will most certainly



(a)



(b)

Fig. 12. Cylindrical $D = 2\lambda$, $L = 10\lambda$ PEC cavity. Monostatic RCS versus azimuth angle θ at 10 GHz. Comparison with FE-BI results with PEC ground plane from Liu and Jin [8]. (a) Horizontal polarization. (b) Vertical polarization.

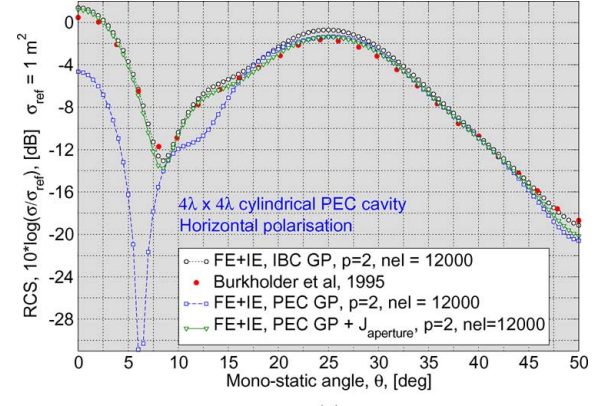
improve the IBC model predictions for grazing illumination. Provision of a half-space Green's function for the absorbing case will make the far-field determination for the absorbing screen model exact in the same sense as it is for the PEC GP model. In other words, it will no longer rely on Kirchhoff's approximation. More accurate modifications of the PEC GP model probably compromise the computational efficiency. Finally, our cavity RCS predictions are in excellent agreement with other published numerical results and are shown to be reasonably converged.

APPENDIX A

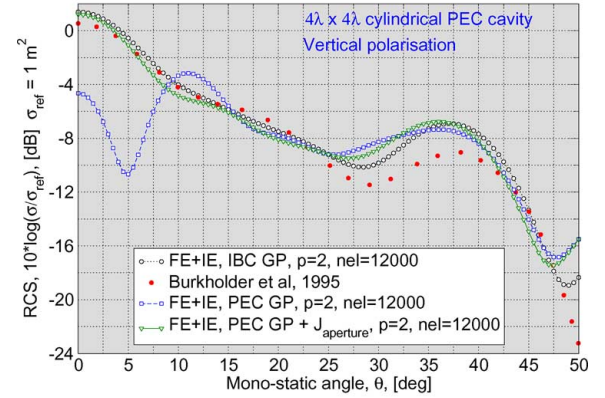
SUPPLEMENTARY EXAMPLE VERIFYING THE PEC GP MODEL IMPLEMENTATION

As a supplementary verifying example we use a cylindrical cavity with a diameter $D = 2\lambda$ and length $L = 10\lambda$ considered by Liu and Jin [8]. The cylinder has PEC walls and a flat PEC bottom. Our hexahedral element mesh consists of totally 8400 elements for $\Omega \cup \Omega_S$. Uniform $p = 3$ and $p = 4$ approximations yield 738.959 and 1.713.118 degrees-of-freedom, respectively. Monostatic RCS is computed from $\theta = 0$ to 90 degrees with step-size $\Delta\theta = 1$ degrees.

Fig. 12 shows that the RCS GP cavity RCS predictions obtained with our FE+IE formulation are in very close agreement with those obtained with the 3rd order FE+BI method in [8]. Furthermore, the virtually overlapping results for uniform $p = 3$



(a)



(b)

Fig. 13. Monostatic RCS versus azimuth angle θ for a $D = 4\lambda$, $L = 4\lambda$ cylindrical PEC cavity. Verification of IBC ground plane model predictions using iterative physical optics model results Obelleiro [19]. (a) Horizontal polarization. (b) Vertical polarization.

and $p = 4$ extensions in Fig. 12(a) show that our results are converged.

APPENDIX B

SUPPLEMENTARY EXAMPLE VERIFYING THE IBC GP MODEL IMPLEMENTATION

Using the $4\lambda \times 4\lambda$ PEC cylindrical cavity studied in [19] and [20] we can compare our RCS predictions for the infinite ground plane model and the absorbing screen model with those obtained using an iterative physical optics method [19], paired with the Kirchhoff aperture integration procedure. To our understanding, the analyses in [19], [20] consider only the interior cavity problem, i.e., the exterior problem is not taken into account other than through the near to far-field transformation embodied in the Kirchhoff aperture integration procedure. In Fig. 13, the monostatic RCS is plotted versus the azimuth angle, for the horizontal and vertical polarizations, respectively. We observe that our uniform $p = 2$ discretization with 12.000 hexahedral elements in $\Omega \cup \Omega_S$ with the absorbing screen model, produces RCS predictions that are close to those obtained by Obelleiro *et al.* [19] with the IPO and the Kirchhoff aperture integration procedure. Here, we note that the infinite PEC ground plane model significantly underestimates the RCS for angles close to axial incidence. It should also be noted that the IPO technique is a high frequency approximate method.

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